

ActInf MathStream 011.1 ~ "Structured Active Inference", Toby St Clere Smithe

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hello and welcome it is August 12th 2024 and we are in active math stream number 111111111 and with repeat friend Toby SMI today we're going to talk about structured active inference there will be a presentation followed by a discussion so thank you Toby for joining and everybody for watching and asking questions and looking forward to learning more here so go for it cool great thanks Daniel thanks for having me on again um so as as Daniel says this is going to be a talk about structured Act of inference which is really um my account of the structure of active inference systems and really it kind of extends Beyond Act of inference um to any kind of system that you could think of as as being kind of agent and that has some kind of compositional structure in particular but we'll get we'll get to that okay um the way I like to begin thinking about structure of and act inference is at the moment it's it's kind of like if you wanted to model in active inference uh an agent or a person getting onto a bicycle or getting into a car then I think as things stand it might be a bit tricky because that process involves like changing the agent's Mark of blanket or how I like to think of it is changing the agent's interface and to be able to talk about that precisely you need to have a good notion of what interface is so you know you as you're you know getting into the getting onto the bike let's say you move you change from you know walking by taking steps one after another to you know um like change your direction by like leaning and and pedaling so that kind of type of actions you can take and the things that you pay attention to change yes because you're now this kind of composite system you and the bicycle and it's a similar story when you get into the car um so that that really that means that you don't just have like one like set of observations and like one set of actions you can take this this kind of composing process means that you have to have a bit more flexibility and in particular you need to say precisely what those sets of observations and sets of actions should be so the basic idea behind struct structured active inference is really just to put the ideas of active inference into categorical systems theory and so categorical systems theory is just an account of how dynamical systems of very general

kind um yeah are made up of parts and how those parts composed together and it separates out the notion of the system from the notion of its interface and so this allows us to be very precise about what is the kind of agent markup blanket and how does that change so the we inherit kind of number of features of categorical systems theory in this notion of structured active inference in particular we inherit this kind of idea of structured interface and that means not only do we have to be precise about what the interfaces are but we also are able to say what the structure in some sense of the interface is and how does it change and how does it you know interact with other kinds of interface I'll say a lot more about that in the next few slides um because categorical systems theory is all about the composition of systems and their interfaces so is structured active inference and that means that we get a a nice kind of um modular account of active inference and we can compare active inference with other accounts of agency and we can compare active inference systems of various different kinds in a nice precise way and so that means we could sort of uh translate between continuous Time and discret Time systems or various different kinds of continuous time or stochastic systems those are all different systems theories of their own and active inference has lots to say about how to do um inference and policy selection on those different kinds of systems but C categorical systems theory means that we're very precise about what what those different systems and how that inference process works and so this this kind of comparability comes into um is is formalized by the notion of morphism of systems so this these morphisms are they come in s of two kinds in systems theory we get morphisms of interface um those are the things that allow you to sort of plug yourself into another sort of bigger system in some sense that is as you get onto the car you sort of get onto the bicycle you're sort of wiring yourself into this composite human bicycle system but they also these morphisms they also Encompass changing the model that is changing the model of system but also changing an individual model itself so doing things like structure learning parameter change so normal learning um you know meta learning or things that we might think of in Neuroscience as effective connectivity these are all kinds of morphism of system and so this all of this stuff comes out of packaging active inference into this categorical systems theory language so with this we gain a lot of things not only do we sort inherit these features but we gain sort of some nice new capabilities as well so I've already talked about what we get from structured interfaces um so that means that we get systems where you can like the the kinds of actions you take might depend on the situation or context or mode that you're in or that can act on very sort of precisely defined structured interfaces like computer apis um we get Notions of agents that

manage agents in a precise compositional way or sort of meta agents that can use active inference itself to change their structure there seems to be an interesting link between um structure learning as in within one model and and planning as in choosing your sort of policy your series of actions both of those seem to be about composing sequences of morphisms and another thing that comes out of um putting active inference into the language of categorical systems theory is that we can also make contact with categorical logic which can also be applied to categorical systems theory and that means that we can do things like or you know we in principle can do things like specify systems goals as logical predicates um talk about uh contracts that might have to hold between agents or um or sort of uh on their behaviors that you might you might something you might care about if you're interested in safety of artificial intelligences um and this logic because it's in this systems theoretic framework has to be compatible with the patterns of interaction of the systems now I'm not going to say very much about this because it's really quite a big and open problem like spelling out all of the details here but I've got a couple of slides at the end about it um and in in that same kind of vein and similar to this notion of um policies as kind of being sequences of morphisms um structured Act of inference because we uh give some kind of typing or context information to the actions that are available that means that we can do things like type checking in principle on policies and that's a kind of rudimentary way to verify the safety of a policy I.E just that it makes sense so the basic um idea of categorical systems theory um this is the math stream so I'm going to be quite mathematical here um so there are two ingredients as I mentioned one is that we have a category of interfaces so that is just something something very syntactic it says okay here are the the interfaces and Mark of blankets of all my different systems um and and that those those interfaces form the object of this category and then here are all the ways that these in interfaces could sort of link up or connect and these these um wirings um they become the morphisms of this category of interfaces um but there's information in the category of interfaces about the systems themselves it's really just a sort of syntactic gadget for talking about how things connect up and so to to attach systems to interfaces we have what's known as an index category so we have for each possible interface there's a whole category of systems that might somehow like fill that interface with a system so we think of the interface as like a box and then the the index category of systems over that category of interfaces fills that box with an actual system it sort of brings it to life and it being indexed category means that this um these categories of systems over interfaces have to be compatible with the wiring up so if you have an interface one box and another box side by side and you connect them together

using one of the morphisms you're also able to connect the systems in in this way compatibly that is you know once you've plugged your boxes together the system sort of interacting this way accordingly um so that allows us to do this kind of um connection of systems um it's not normally just a category we also typically have what's known as the monoidal structure on the category of interfaces and that means that we can place systems side by side we don't just plug systems like one into another we can have like two systems next to each other and then sort of connect them into a bigger system that is like you know your body is made of lots of lots of cells next to each other but they're all sort of interconnected in and that's that's sort of um um formalized by this monoidal structure which is sometimes called tensor and represented with this symbol here and so the systems theory again has to be compatible with this tensor which means if you put two boxes side by side and systems in them then you also get a system in the sort of composite like double box system and so to do wiring really often what happens is you have one box I've written it as P here and another box written as P' you put them together into this $P \otimes P'$ that means that you can connect these a system over this and a system over this into a system over the joint interface and then say you've got some some wiring so wiring would be a morphism $p \otimes P' \rightarrow Q$ then the systems theory allows you to take that wiring and wire the systems together accordingly and then you that means you've got a system on this like VI interface and so often you see diagrams and systems theory it to a bit like this this is p this is p' this is q and then this is like some like way of wiring the systems together and then this is you know this is your wiring and then you've got the S_y and they you get you get a sort of composite system like this so I've gone very fast this is the basics of categor categorical systems theory it's as I say it's a it's a language for talking about building complex systems from their parts in a nice sort of formal compositional mathematical way and so what we're going to do is put active inference into this language in a way that generalizes active inference in this sort of quite big New Direction that allows for things like mode dependence of actions and so to do that I want to introduce a particular category of interfaces um the category of polynomial functors um you if you know the work of David spivac and his colleagues he's a big proponent of this category for talking about things like systems and Dynamics and so I'm going to adopt his notation um in talk about this category um he calls it poly and there's a nice book about this by David and um Nelson new that you can read it's freely available online so a polynomial it's really again as I say it's like a syntactic gadget for talking about the interface of a system and it has kind of two key ingredients and it's really just like a polynomial that you see in high school algebra so it has like a set of

summand which are these like white the something things and they're indexed by this P of one set this is a set and each each sum and has another set attached to it which is the exponent so it's really just like you know $y^2 + 3y + 1$ um this is a polynomial and you know we could tell this could be like $\sum_i y_i$ to the x_i if you sort of you know if we think of the three y as like $y + y + Y$ and so this is a sort of General representation of a polynomial and we think of the the indexing set p of 1 as the set of the uh of a set of the systems outputs or the configurations that it could adopt or in active inference language like the observable states of a generative model and then for each possible configuration of the system as I say you have this set which represents the inputs or the actions that a system might take in that corresponding configuration and so that's just what how we think of a polynomial um and then there are some things you can do with polynomial in particular there's one like trivial polynomial one that has just sort of one configuration and one possible input so there's no possible variation in its configurations and no interesting actions that it can take um we can put polynomial like side by side using this tensor of product and this has the kind of expected effect on the Pol uh a configuration of a tensor is like a configuration of each part and an input is or an action is a an action or an input of the corresponding parts like that and mathematically we can translate between this kind of polynomial representation and what's known as a bundle which is just a function so these are all sets the P of I and the P of one and so we can do the co-product or disjoint Union of these sets by summing over the eyes and this means that the elements of this big set here are pairs of an eye an indexing I in P of one and an element of the corresponding uh exponent set and this bundle is just a function that Maps i ix down to I and this is uh useful for set for defining the morphisms of polynomial which will do the things that I think of as wiring and I'll talk about that in a moment but before I do so I want to um introduce one important example of polynomial which are the monomials because they don't have interesting um they don't have many interesting sort of uh non-trivial summand they're just like one term so and we write these as y to the A and these are going to be familiar because they capture the kind of classical Act of inference so here I think of the O as like the set of observations as an of an agent and the exponent a as the set of actions and so these actions are like the inputs to a generative model that makes it transition and the observations are like the sort of the the sort of data that the generative model or the environment produces that the agent receives and this is a polynomial because you're just summing over the a the O 's and for each o you have the same a and the bundle just takes a pair of o a and projects O Okay cool so let me talk about how these things sort of wire up briefly um these this this is done using the morphisms of pols

which are otherwise known as dependent lenses if you know about um functional programming for instance so a morphism is going to represent taking a system and putting inside a bigger system that is like if you've got two small boxes and you wi them together to bigger to build a big box um or you take two cells and you put or take multiple cells and you put them together to sort of make a multi cellular organism so if we've got polinomial p and Q a morphism from P to Q is given by two things um one sort of goes forwards from P to q and one sort of goes backwards and the forward One S of takes the outputs of the inner systems and turns them into an output of the big system and the backwards one takes the inputs to the big system and the outputs of the inner systems and uses them to produce inputs to the small systems so like one of them sort of goes from inside the box to the out that's the the the f one map and then the other one sort of goes in um but it might take into account the sort of internal inputs as well so it can be used for like feedback so that is to say we've got this um outputs map so we take outputs of the inner box to the outer box and for the corresponding inputs in the outer box so that's these y 's we go backwards but we have to account for this like mode dependence which is a bit sort of funky but this just says that if we've got an output corresponding to position or configuration I of the inner box then we're allowed to look at the inputs that correspond to the corresponding output and then we sort of back propagate those to the inputs again in configuration I so there's a bit of sort of accounting um you have to do to keep track of these like the the sort of um of the configurations of all the systems but that just means that you know there's some like mode dependence or some like you know you might have some if you think of the sort of multicellular organism thing then you might have some like you know ion channels which are open or closed on the in on the sort of membranes of the cells and what they do sort of depends on the state of all of those cells and so you have to account for that in these morphisms but in the monomial case where you don't have this like context or configuration dependence um the morphisms become a little bit simpler they just become like a forwards thing so here we've got an A and an S going forwards and we we have to figure out a way of giving an X and in the backwards direction we have an A and an s on the inside and a y as well coming in and we have to figure out a way of giving a b and a t so this this really just encodes the wiring of these two boxes into the bigger box like I said so using this category poly whose morphisms are the dependent lenses and objects of the polinomial we can give a system the which um captures generative models um I'm going to just talk about discrete time ones in the in the traditional Sense on the monomials but also has some kind of more complicated mode dependent structure um that I think could be useful in general as well

so this is going to be an index category like I said Maps the polynomial IE an interface of the kind that I was just talking about to a category whose objects are generative models on that interface so that means that they are going to be a quadruple like this so that is a state space S um I'm just going to think of it as a set a prior on the state space which we could think of as like a goal or just prior beliefs about the agent States um a likelihood which is a sort of stochastic output morphism um in the case that this is um just a monomial $O \times Y \rightarrow A$ then that's just like $P(O \text{ given } s)$ and then there's also this transition map which says okay well given that I am in this state um I sort of can sample an input that has to correspond to the um configuration that I'm in so I sample from my π and then okay I'm allowed to receive an input in this corresponding um at like configuration $P(O \text{ of one})$ which is given by my likelihood and then okay I receive an input and I've got a state and I get a new state so that's like the transition function again in the sort of monomial case this is exactly what we normally have in active inference we have um a state s and an action in A and we get a new state it's just that now we're allowed to have the possible inputs to the system I.E the the possible actions that you could take on this generative model to be dependent on the state that the system or the state that the environment is in according to that model um and so that that captures this idea that you know say if I have my eyes closed I can't like receive um you know visual signals anymore uh or or you know if I open my eyes then I can't so there's a sort of difference in the sort of input signals that I can receive or if I'm sort of plugged into my computer I can't sort of easily get up and walk around because you know I might like have to like unplug a wire or something so there it captures this mode dependence um there are sort of important sort of mathematical bookkeeping duties we have to do which I'm not going to do but it's easy to check that it's like functorial which means that it's indexed in the right way and that it's monoidal which means we can put systems side by side and I've only just told you what the objects of this category $\text{gen of } P$ are but it also has morphisms and these are ways of like comparing systems um that are known in computer science as coalgebra homomorphisms um so this captures the notion of generative model that we're used to in active inference but it allows for a lot more structure um in particular this kind of mode dependent structure and this compositional structure so just to sort of say like what the kind of thing I mean by structured interface um I'm taking an example from computer science because um there are a lot of uses of polynomial Science in pols in computer science and one of them particular this for defining like um domain specific programming languages um and I think that one use of this could be for defining like artificial agents that um behave nicely with respect to sort of computer interfaces and so um we can

encode the data of a domain specific language into a polynomial because say we've got a little calculator language that has like two binary operations like plus and times they each take in an input uh take in two inputs we've got a sort of unary operation it just takes in one thing and it negates it and like we've got some constants like uncountably many that represent the um real numbers then we can encode the data of this as like you know $2y^2 + y + r$ because we've got these two little operations what uh two binary operations you've got two possible inputs one unary operation it's got one input and like these constants which don't take any inputs and then we can you know represent the set of all terms of this language as like a set by applying the polynomial to like some X um the those X terms those X elements represent the inputs and then we could think okay well what would be um a machine that like takes in terms of my little domain specific language and returns values it's it's something like $r y$ to the P of X which is again another polynomial and you can do lots of things like this basically you can encode like any small um programming language or any any sort of um sort of abstract syntax tree into a polynomial um it's not just examples from computer science that are relevant but they have a lot of structure and so I wanted to show off those examples I think you could do a lot of things in biology as well using this language um but it's it's kind of more sophisticated than we can do with just sort of a set of observations than a set of actions another thing that I wanted to mention is that in this context um it's quite natural to think of the actions or um that that an agent takes as morphisms in a category and this is where this typing information or this typing idea comes from um because you know in categories morphisms have to compose and so we can check that the sort of they they they are sort of compatible when before we compose them um and so to give you an idea of how this works we can think of categories as like little State machines so here I've drawn a little category it's got objects x y and Z and it's got these morphisms it's got two morphisms from X to Y um f and f' and you know we've got G from y to z and we can go along F and then along G to go from X to Y to z and that takes us from X to z or we can go you know f to Z blah blah blah but these are like steps or paths in a little space and so you know we go from X to Y and then y to z but it doesn't it sort of doesn't make sense you know if you if we had another object over here that wasn't connected to z we could go over here but then we wouldn't immediately be able to go over Z that's to say like if we had an island um or a peninsula or an island with just one Bridge onto it um we could go onto the island but then to go somewhere else we'd have to go back along the bridge sometimes the act like the places you can go depend on where you are and this encodes that data into the notion of action um and

indeed every category is itself can be turned into a polynomial um the configurations of the polynomial are the objects of the category and then the the sort of inputs to a system would be the morphisms of that category out of each object that is to say all the steps you could take out of each object and then a policy would have to be like a composable sequence of morphisms or you know in active inference you might have something more stochastic so You' have a belief over such sequences but this because you have this typing information it means you can check that your policies make sense like you don't try to go somewhere and then somewhere from that destination that intermediate destination you don't try to go somewhere that is not connected or take an action that doesn't make sense in in some other way um and so we can use this to start to think about um like safety of policies um using the logic that comes with category Theory this isn't something that I've elaborated but I think it's a sort of starting step to talk about um safe active inference systems in a very rigorous way um and I already mentioned that other kinds of um important notion are also morphisms in in structured active inference like changes of interface or changes of structure and that means I think that we can rigorously apply AC of inference to itself to do things like um structure learning or to to choose how you you know like connect yourself into a bigger unit like into an organization um and on that note let me tell you about how in this setting we talk about systems that manage systems um this is a generalization or will become a generalization of what is known in the active inference literature as D for hierarchical active inference um and so the sort of formal Gadget that we use to talk about this is known as the closure of the category of interfaces that is to say it has what's known as an internal H H because morphism is otherwise known as homomorphism um and the category of polynomial and also the category of lenses um has one of these internal homs that is a way to refer to the category within itself um and that internal H has to be given a polynomial form and so the sort of configurations of this polynomial are the from say from P to Q they are the ways of wiring P into Q I.E the set of lenses from P to q and then you know we have to give as well exponents to provide a polynomial and these Pol these exponents are going to be all the sort of data that the input to these um to these sort of P systems if we think of the p as like the inner systems that they require and so they require outputs from p and inputs from q and that's exactly what this like input data is here so if we so we've got a polynomial that represents how P 's Connect into q 's and we can think about what uh generative models would be over such a polynomial that is like um systems that control the wiring of PS into Q 's or as I've done in this example control the wiring of a $P1$ and a $P2$ into q and so what would these systems be well they would observe the outputs of $P1$ and $P2$ and

then the inputs of Q and they would update their states accordingly having made these observations and then they would output ways to connect P_1 and P_2 into q and that really is like what a manager does in an organization or sort of high level system does in a sort of hierarchical structure um and there's a sort of nice fact that I'll use later on that if you have sort of trivial inner system I.E just with this polynomial y then the generative models on this H polynomial are just the same as generative models on the outer polynomial um so we can use this to capture not only these hierarchical systems but just kind of regular systems as well um uh there's an interesting or you know sort of side fact or reason why this is known as a closed structure is that this is the same as something that's known in computer science as currying um you might have seen this that if you have a function that takes like inputs in A and B and produces C like this is $\text{Maps } a \text{ } B \text{ to } f \text{ of } A \text{ and } B$ we could also write this as like a function that maps from a to maps from B to C and that says okay well this Maps a to F of a something which is a map from B to C so this is the same idea it says okay well the like ways of like mapping into this P polynomial are the same as like mapping of the joint systems um it's an important structure that gets used a lot in this context okay so I spent a lot of time talking about structured generative models in particular you know I introduce this hierarchical structure and I talked about um the structure of you know mode dependence in a sort of abstract and general way um but it was always about generative models and I haven't really talked much about like doing um action selection or State inference um which is of course a very important part of active inference and in some senses kind of what active inference is all about um but but those processes fit very naturally into the same framework um there's something that you attach to a generative model active inference always starts from a generative model and then describes the process for providing a policy to control it and so an active INF agent or I think an agent more generally is made up of two things it's made up of the generative model and then something that I call a controller which um basically chooses the policy to provide to the generative model it's a way to sort of take this generative model which is an open system and like s of me sort of render it closed it of takes the observations that are produced by the generative model and then turns them into actions that can go into the generative model and so there's some sort of nice Duality between the genor model and the controller and it's this Duality that allows us to formalize um active inference in a nice compositional way um one way I sort of started to like to think about this and I don't really know much about this subject so I I won't make any claims that it's rigorous or sensible but I think of the sort of generative model as somehow like the consciousness of the agent it

represents how the agent perceives the world and then the policy part is it's sort of the subconscious processes which Drive the agent in that gen to model I mean this is just sort of my my thinking I don't know if it's if it's very sensible um but as I say we can fit all of this into the same framework um by by kind of taking poly and this category of generative models and the home structure and sort of putting them all together into um a sort of nice bundle or nice mix so I have this two category called gen poly um made up of all these ingredients the objects are the interfaces the polinomial um you can think of them as generalized Markoff blankets and then now it's a two category so instead of between one object and another having a set of morphisms now we have a whole category of morphisms so we have both one cells and two cells so it's a two-dimensional structure um but the H category from P to Q is just the systems on the H interface so that is um generative models on the structure interface that sort of takes P systems and turns them into Q systems so they're sort of hierarchical manager systems and then the the two cells are ways of compose of comparing these systems and I haven't really talked much about that and they won't they won't feature much in in this talk and they don't feature in the paper either um but they're important if we want to talk about things like structure learning or logic and then the composition of systems is as I've been saying all along we take a system over over one sort of h polinomial and a system over another H polinomial and then we wire them together so then we have like a sort of high level manager here and a sort of lower level manager and they talk to each other to produce what we could think of as like a hierarchical organization which takes sort of very lowlevel things and turns them into high level things um and yeah so we're out to tens of polinomial and that makes this um gen poly 2 category into a monoidal 2 category so we put systems side by side in the way you'd expect um and it means we can draw um uh an active inference system in this kind of string diagram way that category Theory people like um so the generative model is this is is like something like this it takes in nothing and it produces T things that is produces uh well I guess this is a bir directional thing so it sort of is a a morphism from a trivial object into p and it produces P outputs and takes P inputs and then the policy is like it's dual so it takes in P inputs and it produces what the generative model expects as output and then as I uh may have mentioned that no I didn't mention yet um that this this as I've said before at least in the case of just a single system just on P um the home from y to P is just the same as P so that's good this captures um what I was saying before about generative models on an interface and then the the the the controller is a is a model from P to Y that means it doesn't sort of it produces a sort of closed system when you compose it with p and that is that is to capture this picture that

I drew before where you have like um a box which is the gener model which is open and then this controller which sort of closes it off and that's encoded in this of formality okay and so now we've got this definition the the controller of of a generative model is like being dual to it um we can sort of unwind or decompress what that means um mathematically um so the home polynomial from P to Y has this form it's just something that you can work through if you write down that expression and figure out what it means um the outputs or configurations of this polynomial are what are known as sections of P so a section of a of this P taken as a bundle is a way of of going from the outputs and for each output giving an input that that matches the corresponding configuration so it's it's just a way of sort of making this diagram commute as we would say in category theory that is to say okay we we we're in a an output or a configuration and we say okay well um I need to provide an input and yes that's nice it matches what what configuration I'm in and we think of this as like being the data that the world um supplies to the system so the system uh or like we think of the system as producing outputs the environment takes those outputs and then turns them into inputs and so that's exactly what these sections do um so those sections are the things that are outputted by this policy part and then the inputs to the policy part are the outputs produced by the P system and so as I say it's exactly dual so the outputs are the things that the the inputs that P needs and the inputs are the things that P produces and so we can unravel this in the case of um this monomial interface which is going to give us what we get in classical Act of inference um the output path of one of these systems is going to do action selection ction so it it has a little State space which I just called R here um it takes an observation and it produces an action and this is um exactly what the the sort of action selection part process and active inference does it takes a state and an observation it produces an action that is to be taken in the Gen model and it also has this part which does um State inference or you might see it called the recognition model or it computes the basian posterior over the states so we're in a state and we have an observation and then we get a new state so that that this these these thing these sort of four things these two parts of the the the policy system and then the the transition model and the likelihood in the generative model are basically all of what an active inference system is you might include the prior in that as well I suppose um and and so that means that we've got all the data of an active inference system in this very neat little package it's a generative model and it's dual and that's kind of it um there are obviously some like um ways you can extend this and not that's not something I'm doing in this paper in this talk but basically you can do things like consider trajectories over actions or continuous time models or do lots of other

things like this um but I'm going to try to keep it as simple as I can I know I've introduced a lot of mathematical detail so far and then okay so then there are three further generalizations that kind of crop up pretty immediately from this this um that I think are quite important and I want to just introduce to you all um the first of them is that you know now we've got this home structure um and this nice kind of uh two category we can think about um as I've indicated already think about hierarchical or deep systems and that really corresponds to nesting small systems in bigger systems and that you know formally means going from um generative models with sort of trivial inner boundary to generator models which have a sort of non-trivial inner boundary and now unfortunately I've swapped my direction of my home so it goes from Q to P um the story still holds I've just change p and Q round um and I I'll say in a moment like how this un unwind so you can get a sense of what it does and then we do the same to the policy it just does the same thing in the opposite direction now it's a system from P to q but they're still like dual so the intuition is still they're still good I think um that's one thing we can do um this produces uh a a uh a sort of by category a two category or what we ultimately becomes a double category um where the the arrows are pairs of a system and a controller um and uh of the next thing we can do to this is to think about how do we model in a sort of rigorous way the changes of model changes of interface that I introduced at the beginning um that corresponds to adding yet another dimension to this structure um and that formally encodes changes the model as extra morphisms categorically and that makes them kind of accessible to this formalism um not something that I'm going to talk about today uh because I introduced enough stuff here um just to sort of wet your appetite I think um but um in this formalism even in this sort of just sort of two-dimensional setting because it's quite General because it it it says that active inference is is really made up of two key ingredients a system and a controller um it allows us to ask precisely how active inference relates to other theories of agency or other the system and of course lots of people have thought about the relationship between active inference and other accounts like reinforcement learning um but I think this framework allows us to make those uh relationships very very precise indeed and to say okay well you there is say one might hope uh a particular like possibly Universal way of mapping any reinforcement learning system into an active inference system in a compositional way so that is you could do it for the parts and also for the like complex system and you get something that makes sense um and possibly one could go and and I think it's quite likely that one could also go in the other direction and there would be nice sort of we we might expect like nice formal um results like these two processes were adjoint or something like that um but

I I would like to highlight at this point that active inference I think in this setting is really not just one um theory of agency but really it's a whole family of theories of agency parameterized by things like whether or not you take continuous time or discrete time systems because those of course give you different systems theories or whether you do Planning by sophisticated inference or inductive inference or just one time step or whether you um or whether you think about like multiple time steps or whether you take the expected free energy or the free energy of the expected future so there are all these different variations and they give you like different like Notions of active inference and so probably different theories of agency in this sense um and and I think that's important for like making sense of the whole active inference landscape and in particular the sort of um landscape of theories of agency itself because it means that we can have like a nice rigorous map of all of these things and like connect them from you know from um you know connect one thing into another quite nicely um again I'm not going to say very much about this but I I just wanted to Sag it up that that's something that you get out of making precise these structures so I I mentioned that I wanted to highlight how this captures um deep or hierarchical act of inference um and this this happens in I think quite a a natural way I mean I've introduced an awful lot of formalism here but once you get used to it I think it works okay um that hierarchical act of inference just corresponds to a sequence of composite arrows in this two category that whose arrows are made up of you know a uh a generative model and uh control dra is upside down but it doesn't matter um so in this setting again as I've mentioned the monomial interfaces correspond to the kind of normal Mark of blankets that are just made made up of a set of observations and a set of actions um but we get this hierarchical structure in the one cells of this um two category and if we think of the sort of composition in this two category is basically by nesting then you know we could I sort of draw it schematically like this um I I'm indexing like the higher levels the sort of interior levels of the system as like higher in this set in this context so like $i+$ one is kind of more inside the system than I um often that corresponds to like a higher level of like explanation in active inference um but because there's this Duality between like um the generative model and the the controller or the policy part I don't I don't like to say high level or lowlevel because it gets confusing I prefer to sort of think about how deep you are inside your agent and how so or close you are to the exterior world when I'm talking about the direction of this of this kind of comp of this kind of hierarchy and so to the left we go sort of more interior and to the right we go more exterior and these arrows Point sort of in the exterior Direction like this and so a s of hierarchical active inference system goes from interior to exterior it's a sort of uh it's an arrow

of this kind that unwinds to this the following data it's a state of a likelihood that is you know parameterized by the state and the interior belief about observations so of which you could think of as like high level in some context observations and it produces a prediction about the more exterior observations there's a prior on the more interior actions which again you could possibly think of as high level but this is parameterized by the or conditioned on the state and the sort of interior observations and the exterior belief sort of more exterior actions or the system's belief about the more exterior actions and then a transition model which again depends on the actions taken at this kind of exterior level sort of high level in this case this is the of high level actions and it gives us new State on the basis of the Old State and then there's this other part which is dual which is the controller so the control here I've written the state space is like beliefs over the generative model State space so that says it takes it has this again these three parts so one is like a generative process which takes the sort of exterior observations the signals which impinge on the sort of outer mark of blanket and translates them onto the inner Mark of blankets so goes in the other direction this um there's a sort of policy part which does Action selection so it takes in a belief over the state space and um like a high level or a more interior belief about over the actions and it produces um what we could think of as a more exterior choice of action so that could be like passing down a sort of high level motor signal um into like lowlevel sequence of like muscle contractions because I think of the sort of closer to the exterior systems as like the ones being made of of lots of say cells um rather than the the interior model of myself as just a body with arms that move and then finally there's this component which does Bayesian inference over the states it takes in a state and an exterior observation that is the signals which they impinge on my retina and it updates the internal State and that's you know the part that does like um you know that gives you your posterior Over States this is a lot of data um in the literature it sometimes comes out in a slightly less um or maybe slightly more familiar way or a slightly less detailed way where the you don't have the interior observations um which I think was like sort of beliefs about high level patches of information like the object rather than the sort of individual like Ral gangon data or something like that and if you imagine not having these ois oi plus ones in this model then you wouldn't have like this generative process um and you would only have a likelihood which takes a state to an observation but you still get this like hierarchical action selection process um so it just takes like high level AC like high level beliefs about sort of actions like I'm moving my arm into sort of lower level sort of um signals that I pass to my muscles but I think it's nice to allow for

this um hierarchy in the opposite direction on the observations as well okay so the final thing I want to mention is that um the categorical systems theory lets us talk about um let's us use categorical Logic for talking about um goals and uh behaviors of systems and formally it's quite easy to say what we need to do that it's not something that I've done because actually a lot of details but formally like it's relatively simple all you need is the ability to Define what's known as a vibration and a vibration is just like a collection of things indexed by this other thing so again the sort of index category thing Notions I introduced at the beginning is uh basically equivalent to vibrations but the idea is now that over each system we have a vibration of um of predicates of logical statements about those systems um that it's indexed by the systems and the systems are indexed by interfaces and it's compatible with all the changes of system and changes of interface and things like that um but one notion one sort of simple idea of like logical structure or predicate you might attached to systems is just notion of um goal for a system so these priors on the state space um and I haven't elaborated this um but you know once you've got this we can kind of use the structures of category Theory to do things like Define quantifiers or logical connectives and so sort of build up statements about goals out of sort of sort of more primitive statements and compare systems goals and to say possibly things like oh now if I've got a bunch of systems connected together um maybe that they work together to achieve AI their goal possibly like by some process of consensus um this allows us to talk about that kind of thing in a mathematically rigorous way or now we've got this formal logic to talk about um the goals of systems can we use this to talk about Notions of safety or to guarantee that um the goals will be achieved within certain bounds or um that they won't try to do this why this thing or this other thing which we don't like um I think this is important again as I've mentioned at the beginning for thinking about um safe artificial systems um that's all I wanted to talk about it was a lot um I think we can go back and elaborate on any of this um and uh I'm sure we'll come back to it in the future as well um thank you so much for having me I hope it wasn't too bamboozling um but I I'm you always willing to answer questions and if you have more questions please feel free to get in touch afterwards as well thank you Toby wow okay yeah while people are asking some questions in live chat which I'll read to you could you reiterate how does what you refer to as classic active inference with Markov blanket basian graphical formalism how does that overall literature relate to what you discussed today in terms of Applied category Theory right okay so I think um the very concise way way of putting it let me see if I can go back to a slide where it's um relatively well not too intimidating the concise way putting it is that um the systems in active inference that I like

classical that we're more familiar with um here to for are basically like sub Theory or a subcategory of this whole structure that is spanned we would say in mathematics by these monomial inter spaces um these are the ones that don't have this like complicated um State dependence on the actions they just have a space of possible observations and a space of actions um and in particular if you don't think about like these hierarchical or deep systems then it's relatively simple to see that those system those kind of generative models are captured by this formalism um because you get uh you know on over one of these mon omals the monomial represents the mark of blanket um the uh coefficient here is the observations the exponent is the actions um we have a state space we have a prior we have a likelihood and we have a transition model and that's kind of the whole deal with a generative model um the the sort of policy part the thing that I call the controller excuse me um is is is it kind of s like I said uh it's it's the thing that's dual it it does um let's see if I can go back to the slide it does uh yeah it does Action selection I.E planning and it does Bas in inference um I.E it produces the posterior here R is just like a placeholder usually this at least in the story that I've told in the paper this would be like beliefs Over States D here means distributions um so this is distributions Over States um it's just a way of like packaging all of that up into a kind of nice compositional framework so that we can because you see a lot of you know if you look at complicated hierarchical active inference um pictures they often have like really complicated structure but actually these structures typically have like some kind of common repeating elements um that in principle should just be able to be boxed up and repeated and plugged in in a nice way that is mathematically sound and reusable and kind of uh you know I like category Theory because it's a kind of linger franer and and it means that you don't have like small variations in formalism like what does message passing mean well it means lots of different things lots of different people um in this setting you have you you can be very precise about like what your theory of agency is um what the notion of Mark of blanket is one of the things I often find a bit tricky about um the active inference literature is the notion of Mark of blanket is quite is it's kind of intuitively quite easy to graph um but in this kind of dynamical setting it doesn't quite correspond um to the notion of Marco blanky that originally came out of Statistics um unless you do some kind of Maneuvers that unwind your Dynamics into a big like uh um graphical model and like draw certain boxes around some repeating elements then then you can say that the mark of blanket of the system is the uh the parents children co-parents like you would say in statistics but it's it's kind of not really what the job of the mark of blanket is um not in active inference and not in statistic they sort of play different roles an

active inference the mark blanket is the agent's interface it's like the boundary of the agent how it interacts with the world and I think it's nice to be precise about that it's nice to say that what when we say markco blanket we mean interface and our interfaces come together and they can connect up and they can produce bigger systems with interfaces um and that has a particular grammar to it which is not the same as the the grammar of markup blankets and statistics um it might be in certain cases but if we want to talk about the sort of whole world of different kinds of dynamical systems from these discrete time ones that I was talking about to various different models of continuous time ones then it's it's not it's not so easy to make the connection back to mark of blanket and so I think to some people who come from statistics um this terminology is can be a bit confusing and one of the things that comes out of this structured Act of inference is that it it says hey this is how we can be precise about this um it says okay well we have this n of interface it can be whatever you want as long as it's you know it behaves like we think of as interface and I've just told you about one story of interfaces and and basically one story of systems like over the interface but categorical systems theory is very very general it lets you talk about lots of different kinds of interface lots of different kinds of systems and this is why I think that this story of structured active inference whilst it captures the kinds of active inference systems that we're familiar with it's really a theory of the structure of Agents generally and I think active inference wants to be that um and that's why I think that you know here we might want to expect that there's like tradition there's like um canonical ways of relating other theories of agency to active inference or to a particular flavor of active inference and that we can make those mappings precise precise not just for an individual model or an individual system but for all systems of a particular kind um and and that's the kind of universality that I think we should be aiming for um I can explain what these arrows on this page mean but I don't you know I think you get the idea that like this is to say if you've got an agent it's a system on a controller you can like forget the controller and you've just got a system active inference is a way of saying okay given my system I've got a story about how to attach controllers to it how to do policy uh how to do planning how to do action selection how to do state inference active inference is a recipe to take your generative model and provide you a controller and reinforcement learning is a or the the variants of reinforcement learning are also recipes for doing something similar um so it's a sort of mapping in the other direction so I think the thing to take away is that you know if you're just interested in simple models um with like one set of actions and one set of observations then you don't have to worry about this but if you're interested in building big

complex models of big complex systems like organizations or if you're interested in talking about systems whose interfaces can be dependent or whose like actions might depend on where they are or what they're doing or who can change their interfaces or who can come together to form organizations that themselves are like dynamical and changing then we need stronger mathematical tools for describing those systems and those processes and those kinds of agents and that's what this framework and that's what categorical system Theory gives us awesome so so many great points and things I'll just highlight two pieces that I think really hit you called it a particular grammar that knowingly or unknowingly plays on the particular partition splitting things into particles and being that partitioning and this is also a grammar of particles and it's a particular grammar like there are sentences that are syntactically or grammatically correct so that's this kind of like syntactic level of typing there are certain Expressions within the active inference ontology or grammar that would be syntactically valid or not or would be examples of an instance or an existential claim or have these other kinds of statuses and then also there's this sort of conceptual connection that is the semantics that separates from the strong typing of the syntax so it it brings up a lot of linguistic and then also you said it's a recipe for for how to do action and that is like if we were to do dynamical systems modeling of a predator prey system or of temperature in a room but then to bring in the upstairs control section can open up multiple possible like cascading State space ramifications especially if you allow for the heading in and out of the nested system that opens a lot of dependencies that at least need to be accounted for or just go with a special case where it's nothing other than just like a fixed situation which is kind of where the buck stops ultimately for any hyper hyper hyper prior yeah I I I like I like those Reflections so I think um yeah there's a this like particles idea you know I Tred to sort of capture very feeble way I'm not very good at drawing like you've got this kind of nesting of systems um and I'm trying to capture that in this formalism um so you could think of like these like outer the outer boundary as being like as I say your body's boundary or like the boundary of like a whole organization or Society or something like that or ecosystem and then the the sort of outer like the job of the outer system is like to pass messages into the inner parts which are like your organs or the sort of units of a organization or departments or something like that or sort of lower so we often think of a lower level but like they're more like smaller scale or more like Interior Systems um one thing that's not so easy is to talk about like the sort of birth and death of these things I haven't really thought much about like how systems are like created from others but I'm I'm trying to like get at least some framework for talking about like what it means to be one um

before we talk about how to like split them up um you said some things about grammar um yes and I think it's it's I'm trying to use this very sort of um in some sense quite like sparse mathematical grammar of category Theory um but I've tried to do so in a way that as I say is like is quite is quite General and it kind of ends up being inherently kind of you know a bit kind of alien um but I think the the thing you gain from that is this kind of um you know ability to talk about lots of different kinds of thing um and hopefully um you this is just the beginning of this project that it becomes you know you know we get tools and stories for how to you express things in this kind of language um it's it's I think most suited for trying to understand like complex systems um so as I say I think if you're if you're happy with just like normal active inference then it's probably fine great and also I'm wearing an appropriate embroidered onion shirt on the birth and death even though from each nucleuses or cores perspective it is like being at the center of a Bullseye from where it is taking another view might look more like a chocolate chip cookie or like multiple nested organs so it's it's not that it's like just a bullseye even though that's just the two layers that's just the motif of nesting and then when you described how the systems can be put side by side then you could have a situation where it's like the cookie has two chocolate chips and then the next time step there's a third one but it's disconnected so it's the island and then another kind of actor comes in and changes the wiring secondarily so one I wanted to I haven't really talked about at all but one thing I want to be able to talk about and it's kind of related to this question of like when the systems like dissolve into other systems or how can we see things as like uh you know a chocolate chip cookie or a bulla or something like this um or you know how do we you know or a a group of trees or a forest or something um you know this is this this thing it reduces systems in you know it's category Theory so it turns systems into arrows um but really this structure like wants to be what's known as a a double category so here we would the arrows would be like the changes of interface and then we would maybe have like some other structure down here which represents another system and we could then then we have like what are known as squares and these squares would represent how we could see like this like this system up here as a composite or as a different kind of system how we could see the the forest as trees or something like that um so it allows us to represent these kinds of like you know dissolving or these kinds of changes I haven't explained in detail how this works but it's a step towards that kind of process yeah cool and then of course all the great jux positions with more topologically stable systems uh neural networks and then more like the ant nestmates okay I'll ask a question from the Live Chat math for wisdom wrote I wonder if it is possible to

interpret it in the opposite way that the policy is the Consciousness so in that little speculation yeah what what led you to to see or identify one with the other yeah because really I I always found the generative model encapture encapsulates like the world as I perceive it there nothing really that I can do to change like we we think of so the generative model is a system yeah so there's one peculiarity about this which is that um the key ingredient is like a system whose outputs are observations and that seems a bit weird sometimes to me and it may seem weird to people who used to active inference where you could think of the the outputs of an agent truly be the actions but the the the the generative model I the the agent beliefs about the world is a dynamical system which outputs observations so the agent receives and so that that seems like the wrong way around but it it's just how the mathematics requires it to be and so it this thing being so like crucial in the structure says to me that like this is like the sort of this is you know when we see you know of inference text is often like s uh given o and m so it's like conditioned on a particular kind of model this this gamma is M and like this m doesn't typically change an active rence unless it's by some like process of evolution for an individual agent this model is fixed and it this this thing it like represents like everything that I see and it and it it represents how I expect the World to Change given the actions I take and it's like it is in ter say it's the agent's world the agent there in active inference there is no like the agent can't access what the real world is like only has the generative model and so that's why I think this is like this is like my world this is like my conscious experience and then so what actions I take I mean maybe so maybe this includes some self model and I think you know I'm doing planning or I think I'm you know behaving in a way that's reasoned but actually of acting on Instinct according to these subconscious processes I mean I don't know much about Consciousness science but I think that that kind of story is possible and so this stuff seems to be the things that I don't necessarily directly experience in the same way that I directly experience like my visual signals um but they are the things that make me like me they are the things that provide my input to the world um and again yeah that's all like Dynamics and it's maybe stuff that I observe but maybe I only observe it because it forms like an interior part of this model which might have some complex structure of the kind that I was talking about um so that's why I thought of it that way around um and of course like you know maybe there is some deliberative planning process or like you know some internal like reasoning that goes on in my conscious experience and I do have an inner monologue and often it does seem like that so I think that there is some interplay between these but possibly I think that stuff is borne out by having some complex structure in this rather than in these like things that I think of a

subconscious processes like I don't I don't I don't like experience my like own Bas and inference processes except in the cases that they go wrong like or like I have some illusion like sometimes I've gone into sometimes I'm in a building that's particularly symmetrical and I go out of like a a list then I go into one place and I'm on the wrong side and it all feels really weird like but that's that's not like the same kind of cons that's kind of unusual experience anyway that that's that's my feeling there what an unusual experience I I'll read from math wisdom he wrote I think that makes sense and now I understand the distinction better the generative model creates beliefs the policy creates actions but our actions are faded we are only free to choose our beliefs our Consciousness has to do with the beliefs we take our will did I understand the distinction correctly I think so I think that sounds like a good good way of putting it I mean this is this is really just my basically this is like my armchair philosophizing I don't know I mean it would be nice if this matched like what people think is actually going on but it's just what the sort of mathematics seem to be saying to me at least towards the applied side what kinds of of programming languages or approaches or open- Source software or methods can people use to start exploring this that is a great question and you know what at the moment that doesn't have a great answer the best answer for now is that I mean at least for like modeling is that in julia um the guys at the topos Institute have been producing a collection of packages in the sort of Mia of their project algebraic Julia which is all about representing categorical structures and um doing compositional modeling in in Julia um but Julia itself is not like necessarily well suited to describing things describing the world in this in this particular kind of compositional way it has its nice compositional features but basically in algebra Julia they've written like a little like sublanguage in Julia itself that encodes these categorical structures um one of the things that's quite nice about category theory is that it's very like you know it's about composition and so it's very modular and it gives you a lot of things that you can change um I'll let me go back to where I talked about um systems theory right at the very beginning yeah so you know I just write down this arrow and it maps from like my category of interfaces to the category of categories and in in principle I could just this should be like a little program which takes in which where I just like write I provide it with a polynomial and it gives me back this other data structure um this category of systems um and then I should yeah I could I I should be able to talk about morphisms between these things and I could be a map like this or make like changes to this and have my computer automatically do it um but that's just not possible with any kind of computational tools that we have right now um and I often feel that it should be possible to do this in a way that's not necessarily like tied

to a single programming language like like Julia is to Julia there are also like dependently type programming languages for like proof assistance but if you write in those then you're stuck in that world one of the things that's nice about like the worldwide web is that these days you can write into lots of languages and they get compiled to web assembly and we can interact in a sort of language agnostic way and this this mathematics should be language agnostic but we don't have any languages or tools for talk about it yet um it's something that I think should change uh people who are interested in Chang here it should like talk to the topest people talk to me and I would like to see it I would like to see tools which are really nicely compositional and people as I say the topest people are working on it they're building other things in Rust um those are the best answers I know of for now but they're not like Optimal thank you well it's an optimal agenda so that's very cool and to be able to connect even potentially in the coming months between the algebraic Julia and RX INF fur. JL could already off for some cool opportunities yes I think that would be a great project cool um over the last years like working on this from your dissertation which we discussed in the live stream 54 series all the other fun and learnings that we've had um how have you seen like the process of going about that recipe evolve from like the chapter 6 in the 2022 textbook how do you feel like as more and more of the applied category Theory developments are formalized and then also like operationalized how is that trajectory of like what it looks like to apply active inference yeah I think it's it's been very interesting and very exciting I would say for a long time I I was also like quite I guess relatively confused about things um in some way and this is a thing that like this is um something that I haven't yet been able to resolve uh and and I'm sure some people will know how it could be resolved um but I'm just not yeah one of them which is that active inference has come out of a sort of free energy principle world where the idea is that actions are selected in like the same kind of way that you do perceptual inference by like minimizing surprise or free energy um but the sort of formal yeah so there the formal structure of uh like pure perceptual inference is it has a similar feeling right so it has something that goes from like it f like inside to outside and then something that like goes back this is like the thing that does prediction or and then this is the thing that like does you inference um and so this structure of these two things you know you by no surprise obviously has this similar bidirectional feel but I haven't yet shown that this action selection process is exactly of the same type as Basin inference of perception and I would like it to be it's it's kind of frustrating to me that so I thought for a long time how do I get action when I was writing my thesis how do I get action into this um and that's what this project really has been about it's

like getting action into the story finally um but the relationship between them isn't as neat as I would like and so um if anybody wants to talk about that I'm always interested in trying to like show that like action selection really is a kind of like surprise minimization or approximate inference um so far it's only kind of vaguely in the structure um and this reminds me of something that I wanted to say in response to your remarks at the end when when I just finished Daniel um that I've said here roughly like the data of an active inference like a hierarchical active inference system it has these components it's got like a likelihood model a transition model A you know policy thing a state inference thing and you know maybe it's got this like generative process part and this action prior part as well um but this is all just structure this is structured active inference this all just about structure none of this yet is about like how do you actually choose this likelihood or how do you actually do like action selection or how do you actually do state inference um the stuff that I was doing in the work that I was doing in my thesis was about various different ways that we actually do inference like particularly um inference in the brain using the variational inference under the llast approximation um but what one want well like what I want at least is to say that act of inference or a particular flavor of active inference is like in some sense Universal and that means that there is like a canonical choice for a given system so like we assume that somebody comes along and gives you the data of your generative model it could be you Daniel or it could be like Evolution science or something like that and then it says okay well active says there's a canonical way of choosing actions that is like as optimal as you can be and there's a canonical way of doing inference that is as optimal as you can be um and this story that I've told doesn't is agnostic to that um I'm you know a lot of people have spent time saying here are proposals um this framework although being agnostic allows you to State like what are known in category Theory as universal properties and so those are ways of saying that this is in some sense canonical or optimal in some like mathematically precise way and it would be nice to like have the those stories about the universality of active inference I the ways of spelling out these functions um to be expressed in that language um yeah so so yeah that's that's my answer two answers one was about this annoying thing with inference and here about the universality yeah there's so much to go into and you mentioned like duel a lot and the interface can be seen it's like two onions where you could peel off the interface and Associate it with one but that would be a symmetry break because you could also see it as being peeled off with the other because it's an interface and this gets into like systems engineering what what is the interface and uh it's very interesting topic so I'll see if anyone

else has any final questions like what are your next steps on the research how do you see it being applied and continued okay right so my next steps are really to produce uh longer or more detailed paper about this than the abstract that is available at the moment um in particular the Abra that's available at the moment it doesn't really spell out uh some of this stuff uh in particularly nice detail like um to keep things as simple as I could I didn't really explain like how you can have this kind of um mode dependence transition model as as that that is stochastic as well as having uh likelihoods that are sarcastic for technical reasons it's a lot easier if you just have the outputs of your systems be deterministic this notation right like sum with a sampling is is not something that is um I I think um conical or widely known so I'd have to spell that out so there are lots of like mathematical details but that's like the first thing um and the second thing I think is to get people using this framework um actually talk about building complex systems or actually building tools for building complex system using it um and relating those to you know the the tools that people already use for doing that kind of thing like um PDP or just like you said Daniel RX and th um it would be great to like actually turn this into something that people can use rather than just the mathematical Theory I like this although it's kind of um often people think it's quite an alien language uh because it's unfamiliar but it is in some ways quite ergonomic it it's quite demanding because it because to like prove that something is an index category you have to check all of these annoying conditions and by God they are often really really annoying and I would love you know as a test case for this to build like uh an artificial proof assistant or like system that can help me do these checks because as I said like doing a doing policy like action selection um is very much like choosing a series of morphisms and it's well known in mathematics that proof is also choosing a series of morphisms so I'm pretty sure that we can use this for doing things like proof in a way that's nicer than large language models um so once we've got tools I want to like actually start applying them to things um complicated structured models that you know are probably beyond the systems that we have right now okay the Toby signal has gone out um I'll ask one more question from the Chat math wisdom wrote Toby it sounds like you want to set up a free funter so that suggests considering the adjoint funter what is it that you want to forget if he could Define a funter what he wants to forget then mathematically that could yield the funter that freely optimally constructs the canonical way to choose the action or do the inference yeah what do I want to forget in some sense what I want to forget is like all the stuff that I might have learned that confuses me um but the thing is it's very hard to distinguish the stuff that is confusing from the stuff that is not confusing until you've done all the work so it

may be the case that you have to construct that adjoint in order to see that you've got the forgetful function in the first place U and that's unfortunately I think the situation that I'm in um part a lot of this process I think has really been about um figuring out the essence of these systems like in before it was like approximate inference in the brain and here it's like the essence sort the structure of active inference um really distilling it down into something quite abstract um and really just to say something about forgetful functors um this uh where has it gone yeah this Arrow here that goes from Agents the category of Agents which has these birational general model policy things as it's morphisms and just goes down to systems this is a forgetful functor it forgets the controller the thing that is the policy and um possibly hopefully the the way to see active inference as universal is to say that it's like um adjoint in a particular way or at least it has like at least it is a section we would say of this thing um so yeah it's true what I'm trying to do is find out what those adjoints are um but it's it's hard to know what to forget until you forgotten it and then it's gone I don't know you don't know what you got till it's gone then there's the reconstructed the reconstructed trace and our Niche is becoming increasingly capable from a perception and action standpoint which is bringing up so many fascinating topics so do you have any final comments not today I'm sure there'll be other opportunities Daniel I hope so at least I do too it's been a pleasure thank you Toby this was an awesome presentation it's like big structural Evolution and development in active inference and so there's so many pieces for people to to pick up and and work in all ways developing like the background needed to get to some of these Cliffs and then the better structures to jump off them with yeah please everybody get in touch if you have questions or projects in mind I'm always interested to hear them cool and I hope that we can like reactivate some category Theory ACM in the RX and fur project or in some other Standalone project I'm sure it'll happen so thank you cool peace Toby thanks everybody in the live chat Cheers Cheers e